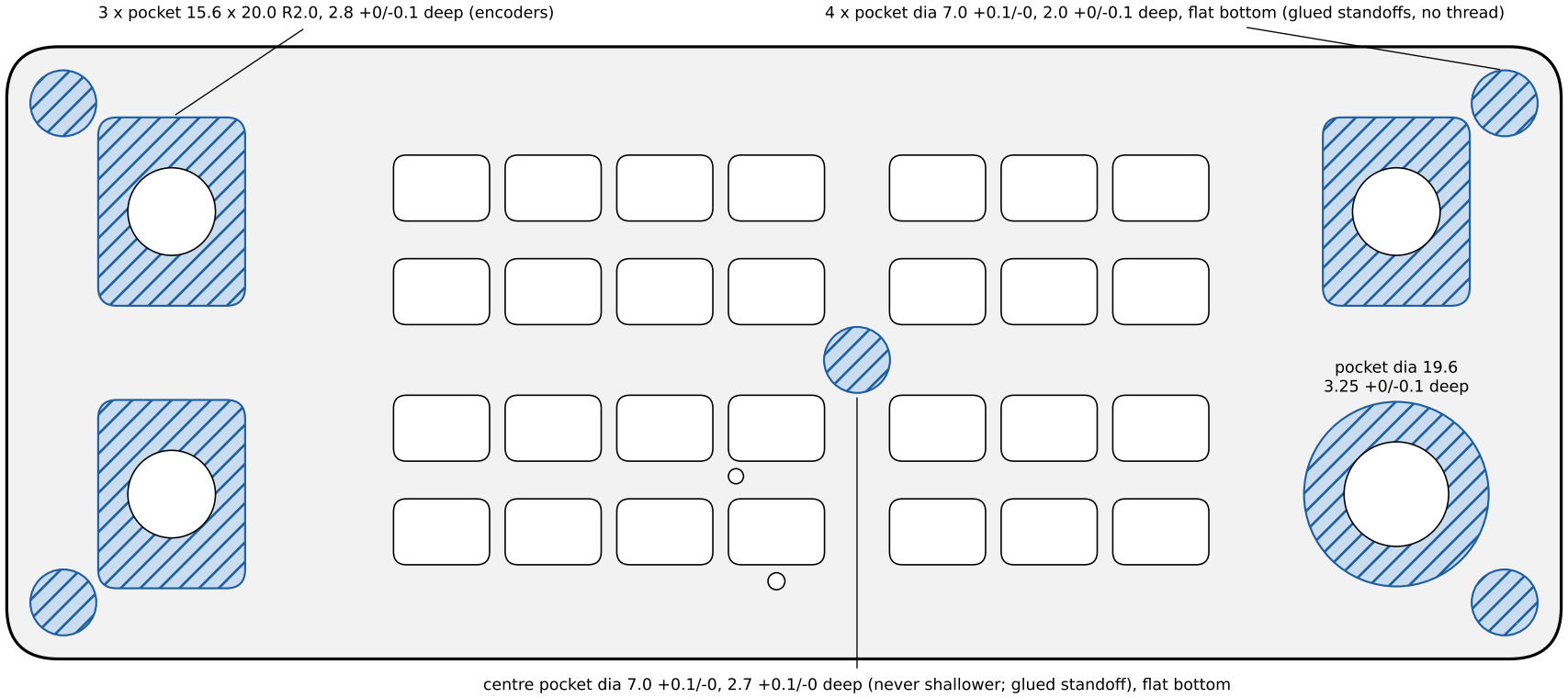
Avionics Panel rev 1 front plate, seen from the BACK (machined side)



Features

key hole	10.2 x 7.0 +0.1/-0 R1.4	through	x28
encoder hole	dia 9.3	through	x3
encoder pocket	15.6 x 20.0 R2.0	2.8 +0/-0.1 deep from back, floor corner R0.3 max	x3
RANGE hole	dia 11.1	through	x1
RANGE pocket	dia 19.6	3.25 +0/-0.1 deep from back, floor corner R0.3 max	x1
SHIFT light hole	dia 1.8 +0.1/-0	through	x1
UNIT light hole	dia 1.6 +0.1/-0	through	x1
standoff pocket	dia 7.0 +0.1/-0, flat bottom	2.0 +0/-0.1 deep from back	x4
centre pocket	dia 7.0 +0.1/-0, flat bottom	2.7 +0.1/-0 deep from back (never shallower)	x1

Notes

- Material 6061 aluminium, 4.0 +-0.1 mm thick before anodising. Outline 165.0 x 65.0 mm, corner R5.5. Volume 30.7 cm3. Units mm, general tolerance ISO 2768-m; flatness 0.1 over the whole plate after bead blasting and anodising (held at its corners, LEDs close under its back in the middle).
- Front face (+Z in the STEP) is flat: no pockets or fastener holes on the front, and no chamfers beyond the edge break below.
- Positions of the 5 standoff pockets and all through holes: +-0.1 to each other (the key caps and the board must line up).
- NO THREADS. 4 x round pockets dia 7.0 +0.1/-0 from the back, flat bottom (end mill, no drill point), 2.0 +0/-0.1 mm deep; metal standoffs are glued into them later. These and the encoder and RANGE pockets: depths +0, never deeper: at least 0.65 mm of material must stay to the front face under every pocket; nothing may break through or mark the front face.
- Centre pocket (between the key blocks) dia 7.0 +0.1/-0 like the corner pockets, but 2.7 +0.1/-0 mm deep: the one pocket that must never be shallower (a standoff is glued in it without touching its floor), flat bottom. At least 1.10 mm stays to the front face.
- Encoder pockets and RANGE pocket: floor corner (inside radius between wall and floor) at most 0.3 mm; the parts' top corners come within about 0.1 mm of the pocket walls or floors.
- Finish: black anodised, matte (bead blasted before anodising); the pockets may be anodised or bare, but all pockets alike (all anodised or all bare).
- Marking on the front per avionics-panel-rev1-plate.dxf, same origin (bottom-left, seen from the front): layer LASER = laser marking, layer UV_AMBER = UV print in amber, matched to #d9a441 (RGB 217, 164, 65, about Pantone 7563 C, roughly RAL 1004) if possible, otherwise the standard orange. Layer CUT = outline and through holes for reference. DXF units mm; fill closed contours even-odd (letter counters are separate contours).
- Key holes 10.2 x 7.0 +0.1/-0 after anodising (never smaller: the key caps centre the board in them). Light holes +0.1/-0 (never smaller: light pipes are glued into them). Deburr all edges: edge break at most 0.2 on the front outline and hole edges; on the back edges of the key holes 0.1 to 0.2 (a lead-in for the key caps).